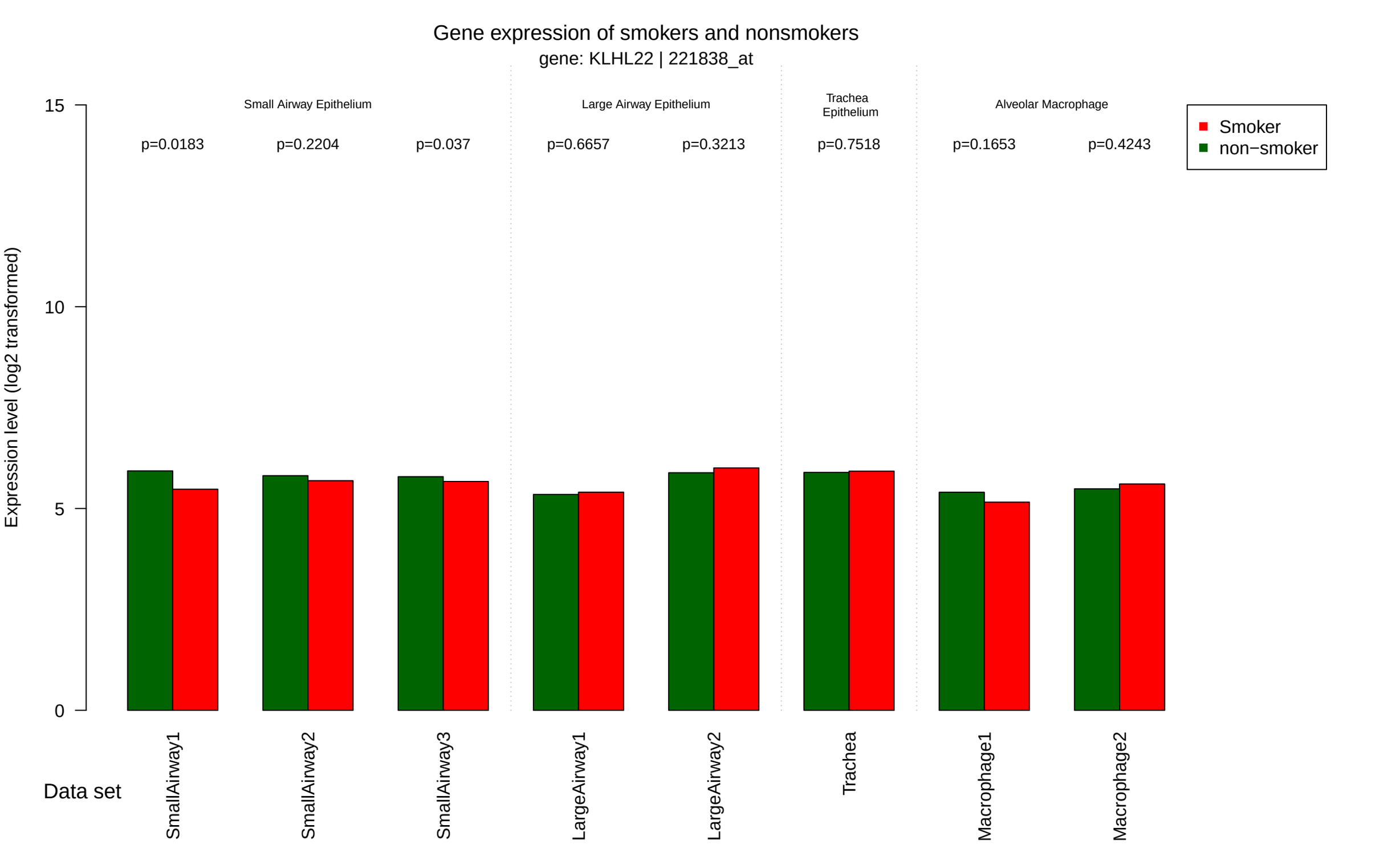
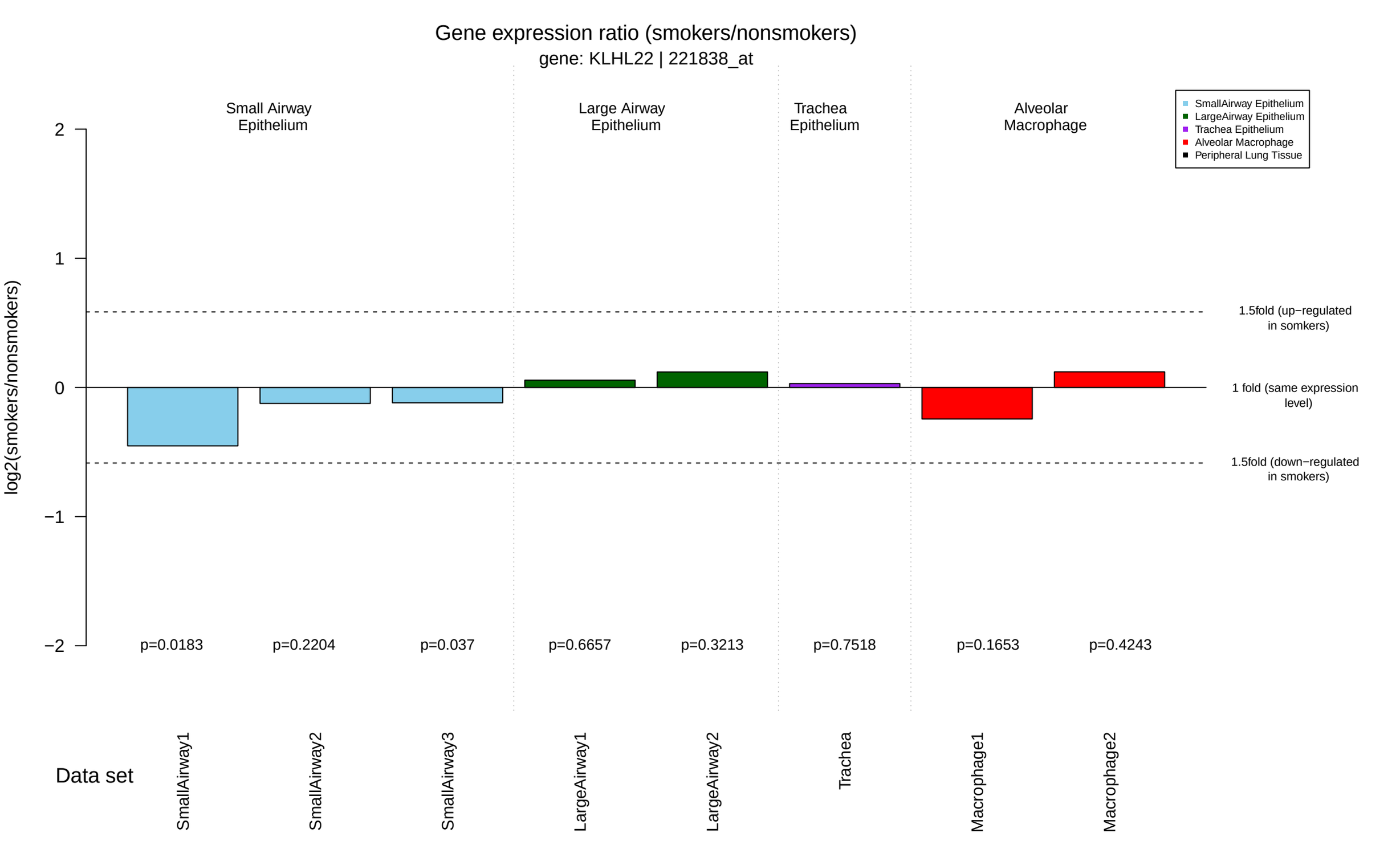
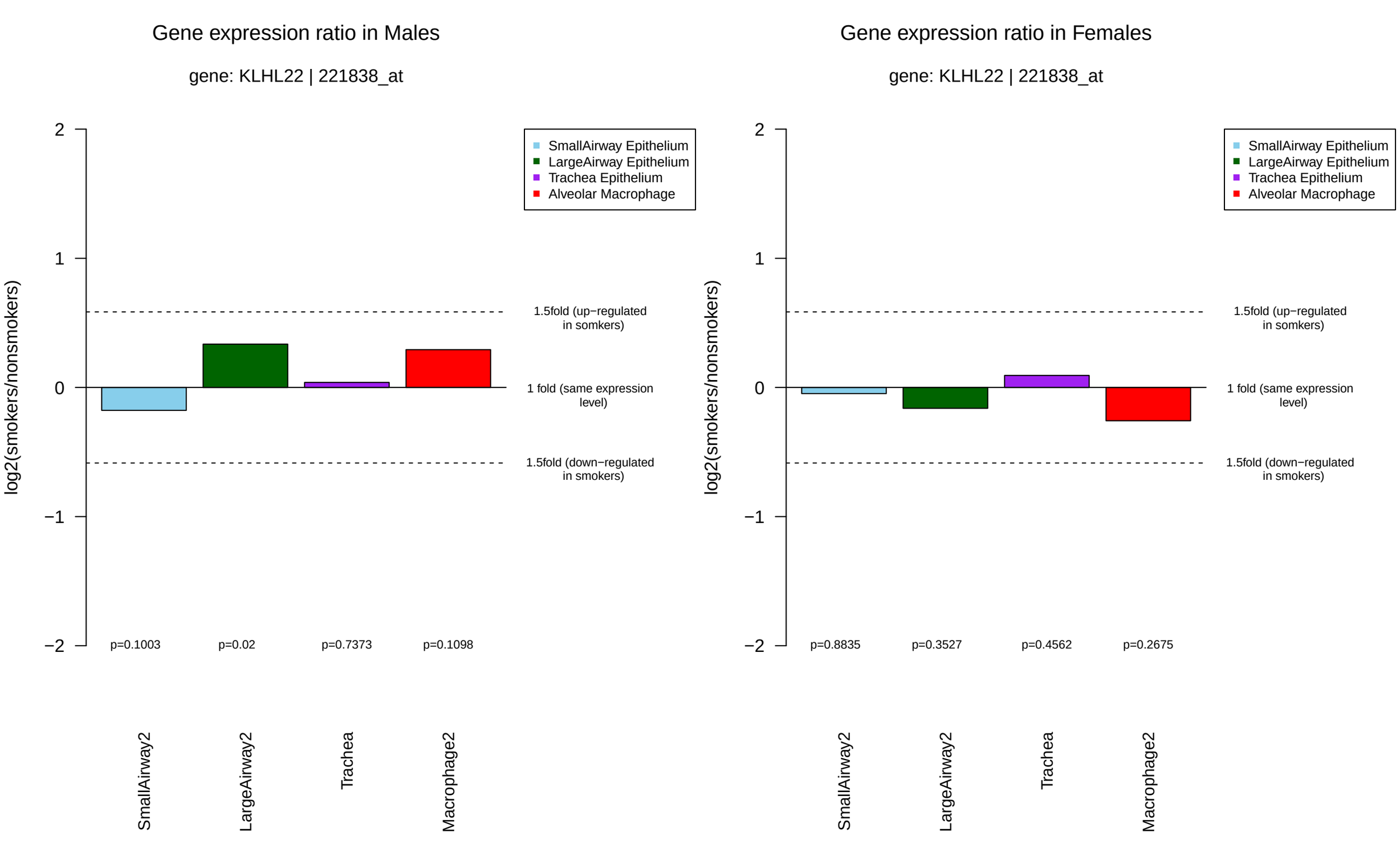
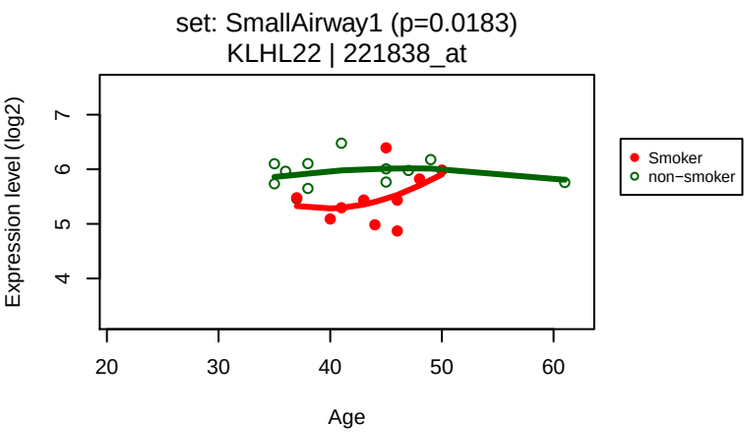
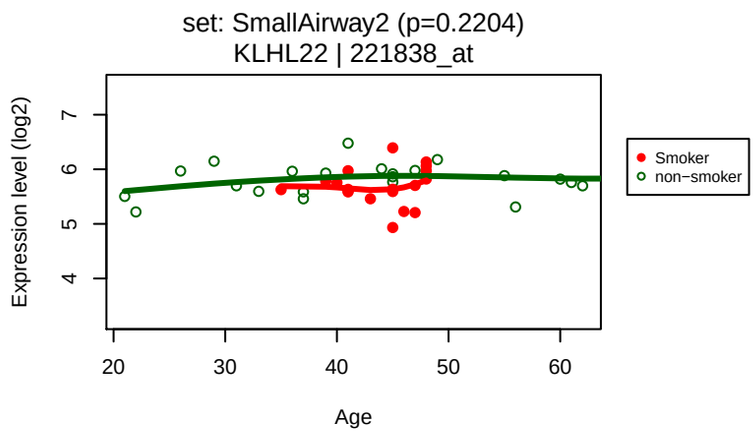
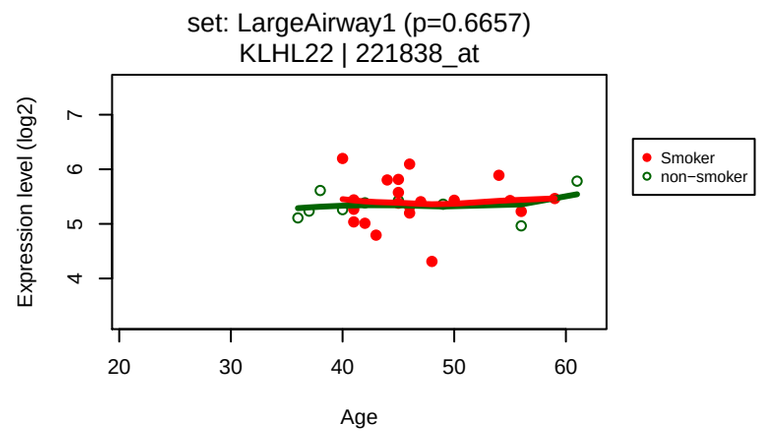

Gene expression of smokers and nonsmokers
gene: KLHL22 | 221838_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0183
p=0.2204
p=0.037
p=0.6657
p=0.3213
p=0.7518
p=0.1653
p=0.4243
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

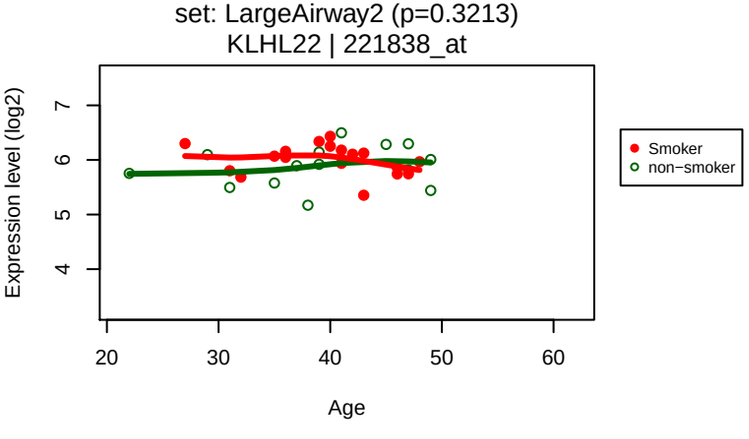

Gene expression ratio (smokers/nonsmokers)
gene: KLHL22 | 221838_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0183
p=0.2204
p=0.037
p=0.6657
p=0.3213
p=0.7518
p=0.1653
p=0.4243
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

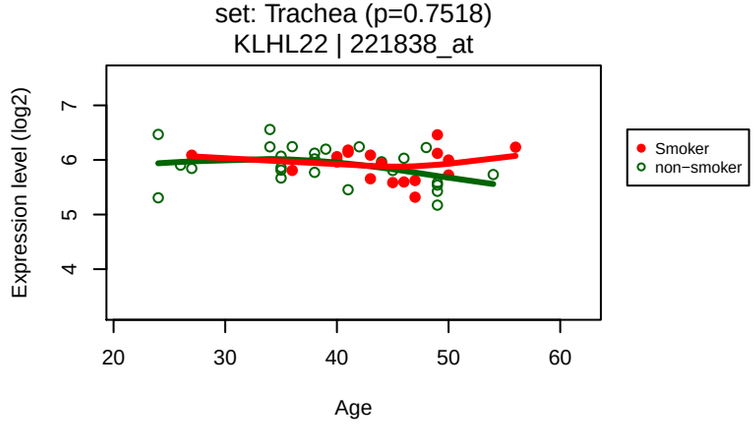

Gene expression ratio in Males
gene: KLHL22 | 221838_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1003
p=0.02
p=0.7373
p=0.1098
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: KLHL22 | 221838_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8835
p=0.3527
p=0.4562
p=0.2675
SmallAirway2
LargeAirway2
Trachea
Macrophage2

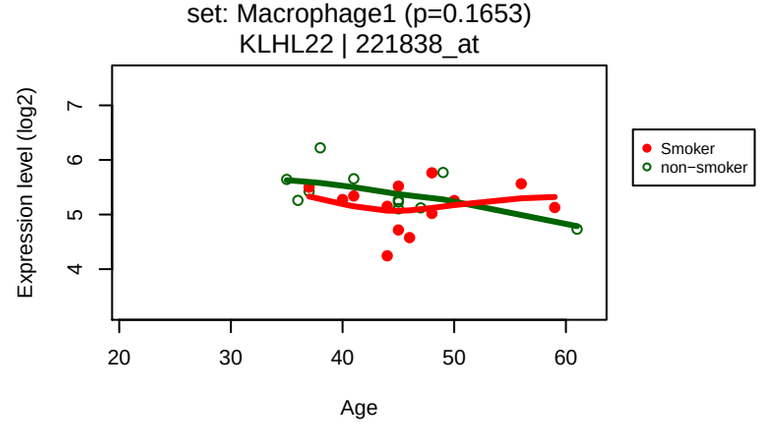

set: SmallAirway1 (p=0.0183)
KLHL22 | 221838_at
Expression level (log2)
Age
Smoker
non−smoker

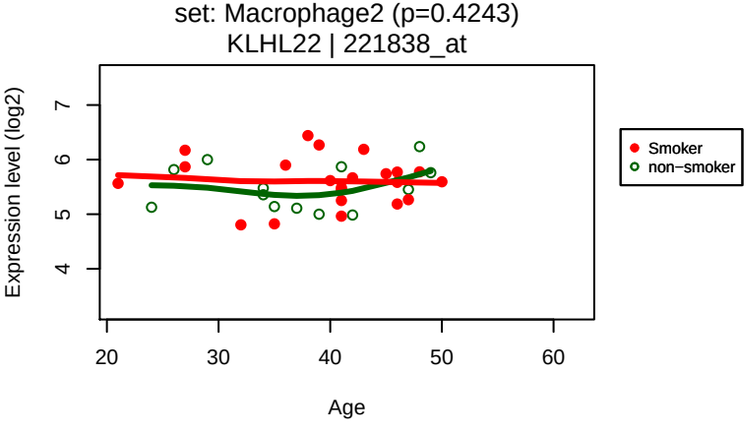

set: SmallAirway2 (p=0.2204)
KLHL22 | 221838_at
Expression level (log2)
Age
Smoker
non−smoker

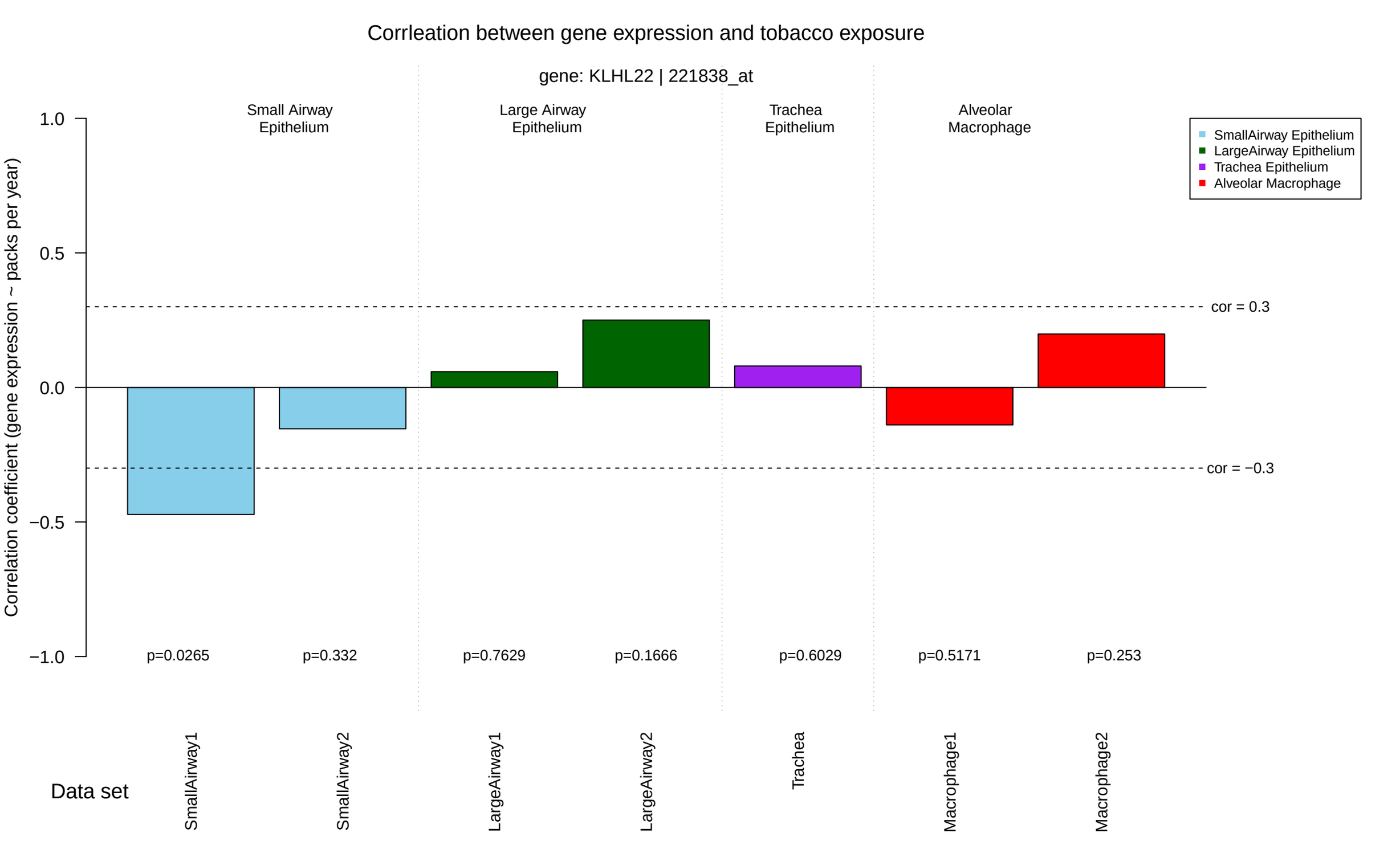

set: LargeAirway1 (p=0.6657)
KLHL22 | 221838_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3213)
KLHL22 | 221838_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7518)
KLHL22 | 221838_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1653)
KLHL22 | 221838_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4243)
KLHL22 | 221838_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: KLHL22 | 221838_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0265
p=0.332
p=0.7629
p=0.1666
p=0.6029
p=0.5171
p=0.253
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2